

Integrable Systems — Exercise Sheet 18

Free fermions: the transverse-field Ising chain

Variant: Questions only

Conventions, as in Lecture 18: $H = -J \sum_n \sigma_n^x \sigma_{n+1}^x - h \sum_n \sigma_n^z$ on a ring of N sites, N even, $J, h \geq 0$; $\Pi = \prod_n \sigma_n^z$; $\sigma^+ = |\uparrow\rangle\langle\downarrow|$, $\sigma^- = |\downarrow\rangle\langle\uparrow|$; Jordan–Wigner $c_n = S_n \sigma_n^+$ with $S_n = \prod_{m<n} \sigma_m^z$, $n_n = c_n^\dagger c_n = \frac{1}{2}(1 - \sigma_n^z)$ and $\Pi = (-1)^{N_f}$. Lecture 18 shows $\sigma_n^x \sigma_{n+1}^x = (c_n^\dagger - c_n)(c_{n+1}^\dagger + c_{n+1})$ with $c_{N+1} = -c_1$ in the even sector ($\Pi = 1$) and $+c_1$ in the odd one; there the momenta run over the NS grid $\pi(2m+1)/N$ and the R grid $2\pi m/N$; $c_n = N^{-1/2} \sum_k e^{ikn} c_k$; $\varepsilon(k) = 2\sqrt{J^2 + h^2 - 2Jh \cos k}$; and every eigenvalue is $\sum_{\text{NS}} \varepsilon(k)(n_k - \frac{1}{2})$ with $\sum n_k$ even, or $\sum_{\text{R}} \varepsilon(k)(n_k - \frac{1}{2})$ with $\sum n_k$ odd, where $e_k = \varepsilon(k)$ except the signed $e_0 = 2(h - J)$, $e_\pi = 2(h + J)$.

Exercise 1 (Jordan–Wigner identities). (i) On one mode, show that $(c + c^\dagger)(1 - 2n) = c^\dagger - c$, $(c^\dagger - c)(1 - 2n) = c^\dagger + c$ and $(1 - 2n)(c^\dagger - c) = -(c^\dagger + c)$. (ii) Show that $\sigma^y = -i(\sigma^+ - \sigma^-)$ and that $\sigma_n^y \sigma_{n+1}^y = -(c_n^\dagger + c_n)(c_{n+1}^\dagger - c_{n+1})$ for $1 \leq n \leq N - 1$. (iii) Show that $\sigma_N^y \sigma_1^y = \Pi (c_N^\dagger + c_N)(c_1^\dagger - c_1)$, and that with $c_{N+1} = \mp c_1$ in the two sectors it has the bulk form of (ii).

Exercise 2 (the XY chain). Lecture 18's reading states that

$$H_{XY} = -J \sum_n \left(\frac{1+\eta}{2} \sigma_n^x \sigma_{n+1}^x + \frac{1-\eta}{2} \sigma_n^y \sigma_{n+1}^y \right) - h \sum_n \sigma_n^z$$

has the spectrum of Lecture 18's theorem with $\varepsilon(k) = 2\sqrt{(h - J \cos k)^2 + \eta^2 J^2 \sin^2 k}$ and e_0, e_π unchanged, and leaves the derivation as an exercise. (i) Show that in each sector H_{XY} is $-J \sum_n [(c_n^\dagger c_{n+1} + c_{n+1}^\dagger c_n) + \eta (c_n^\dagger c_{n+1}^\dagger + c_{n+1} c_n)] + 2h \sum_n n_n - hN$, with $c_{N+1} = \mp c_1$. (ii) Derive $\varepsilon(k)$ and the unpaired energies. (iii) (Computational.) Check the whole spectrum against exact diagonalisation at $N = 8$, and give the lowest level of each sector at $\eta = \frac{1}{2}$, $h = 0.7J$.

Exercise 3 (the ordered doublet). (i) For $f(k) = \sum_m \hat{f}_m e^{imk}$ with $\sum_m |\hat{f}_m| < \infty$, show that $\sum_{k \in \text{NS}} f(k) - \sum_{k \in \text{R}} f(k) = -2N \sum_{j \text{ odd}} \hat{f}_{jN}$. (ii) For $h < J$ the lowest odd level is $E_0^- = -\frac{1}{2} \sum_{\text{R}} \varepsilon(k)$. Compute the splitting $E_0^- - E_0^+$ of the TFIM at $h = 0.4J$ for $N = 8, 10, 12$, and the ratios from N to $N + 2$. (iii) At $N = 8$ and $h = 0.4J$, how far above the ground state is the first level above the doublet?

Exercise 4 (the ground-state energy per site). (i) From $E_0^+ = -\frac{1}{2} \sum_{\text{NS}} \varepsilon(k)$, show that $e_0 = \lim E_0^+/N = -\frac{1}{2\pi} \int_0^\pi \varepsilon(k) dk$, and evaluate it at $h = 0$ and at $J = 0$. (ii) At $h = J$, show that $\varepsilon(k) = 4J |\sin \frac{k}{2}|$ and $e_0 = -4J/\pi$. (iii) Show that e_0 is unchanged by $J \leftrightarrow h$. (iv) For the XX chain ($\eta = 0$ in Exercise 2) with $J > 0$, where does ε vanish?

Exercise 5 (the current). Let $Q_- = \sum_n (\sigma_n^x \sigma_{n+1}^y - \sigma_n^y \sigma_{n+1}^x)$ (Lecture 18's tutorial). (i) Show that $\sigma_n^x \sigma_{n+1}^y - \sigma_n^y \sigma_{n+1}^x = 2i(\sigma_n^+ \sigma_{n+1}^- - \sigma_n^- \sigma_{n+1}^+) = 2i(c_{n+1}^\dagger c_n - c_n^\dagger c_{n+1})$ for $n \leq N - 1$. (ii) Show that in each sector $Q_- = 4 \sum_k \sin k c_k^\dagger c_k$, and deduce that Q_- commutes with H , and with H_{XY} .

Exercise 6 (the open chain at $h = 0$). Drop the boundary bond: $H_{\text{open}} = -J \sum_{n=1}^{N-1} \sigma_n^x \sigma_{n+1}^x - h \sum_{n=1}^N \sigma_n^z$. Let $\alpha_n = c_n + c_n^\dagger$ and $\beta_n = i(c_n - c_n^\dagger)$. (i) Show that $\alpha_n^2 = \beta_n^2 = 1$, that different Majoranas anticommute, and that $H_{\text{open}} = -iJ \sum_{n=1}^{N-1} \beta_n \alpha_{n+1} - ih \sum_n \alpha_n \beta_n$. (ii) At $h = 0$, show

that $f = \frac{1}{2}(\alpha_1 + i\beta_N)$ is a fermion commuting with H_{open} , and deduce that every level is at least doubly degenerate.

Exercise 7 ★. *The edge mode for $0 < h < J$.* Keep the open chain of Exercise 6, now with $0 < h < J$, and let $x = h/J$. (i) Show that $\Gamma = \sum_{n=1}^N x^{n-1} \alpha_n$ satisfies $[H_{\text{open}}, \Gamma] = 2ih x^{N-1} \beta_N$. (ii) Deduce that on the half-infinite chain $\sum_{n \geq 1} x^{n-1} \alpha_n$ commutes with the Hamiltonian, and is normalisable exactly when $h < J$: a zero mode localised at the edge. (iii) (*Computational.*) At $h = 0.4J$ compute the splitting of the two lowest levels of H_{open} for $N = 6, 7, 8$, and compare with $2h x^{N-1} (1 - x^2)$.